Rapid Hamstring/Quadriceps Force Capacity in Male vs. Female Elite Soccer Players

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Abstract

Zebis, MK, Andersen, LL, Ellingsgaard, H, and Aagaard, P. Rapid hamstring/quadriceps force capacity in male vs. female elite soccer players. *J Strength Cond Res* 25(7): 1989–1993, 2011—Knee joint injuries are a serious issue in soccer. The ability to protect the knee from injury depends largely on the strength of the hamstring relatively to the quadriceps, that is, a low hamstring/quadriceps (H/Q) strength ratio is suggested as a risk factor. Although maximal muscle strength (MVC) has often been used to evaluate the H/Q ratio, the ability to rapidly develop force (rate of force development [RFD]) is more relevant in relation to fast dynamic movements. The aim of this study was to introduce and investigate a rapid RFD H/Q strength ratio compared with the traditional MVC H/Q strength ratio in elite soccer players. Twenty-three elite soccer players (11 women, 12 men) performed maximal voluntary static contraction for the hamstring and quadriceps in an isokinetic dynamometer, from which the maximal muscles strength (MVC) and RFD were extracted. Test–retest reliability for the RFD H/Q ratio was high (intraclass correlation coefficient = 0.664–0.933). The initial contraction phase up to 50 milliseconds from the onset of contraction showed a low RFD H/Q ratio compared to the MVC H/Q ratio ($p < 0.001$). These results suggest a reduced potential for knee joint stabilization during the very initial phase of muscle contraction. Two female players—both with a markedly low RFD H/Q ratio, but a normal MVC H/Q ratio, compared with the group mean—sustained ACL rupture at a later occasion. The high reliability of the new RFD H/Q strength ratio indicates that the method is a relevant tool in standardized clinical evaluation of the knee joint agonist–antagonist relationship.

Key Words rate of torque development, gender, hamstring, quadriceps

Introduction

A high incidence of knee injuries has been reported in sports activities such as soccer, handball, basketball, and other ball games (19,22,24). Thus, there is a strong need to assess potential physiological risk indicators to thereby prevent or reduce the occurrence of serious knee injury in athletes involved in high-risk sports.

The contraction force of the quadriceps muscle during knee extension produces substantial anterior directed shear of the tibia relative to the femur at extended joint angles (8,20). This shear can be counteracted not only by the anterior cruciate ligament (ACL) but also by hamstring coactivation (10). Thus, low muscle strength of the hamstrings relative to quadriceps has been proposed to increase the risk of noncontact knee joint injuries (2,3,12).

A hamstring/quadriceps (H/Q) strength ratio based on peak force values during a maximal voluntary contraction (MVC) of either static or dynamic character has traditionally been used to describe the potential for knee joint stabilization (2–4,13). However, a discrepancy exists between the time-wise nature of H/Q strength ratio based on peak force and the potential to stabilize the knee joint during rapid match play situations. Although peak force during an MVC is reached in approximately 500 milliseconds from onset of contraction (1), <50 milliseconds is often available to stabilize the knee joint during rapid ground contact situations (18). Thus, the estimated time of ACL injury ranges from 17 to 50 milliseconds after initial ground contact (18). Therefore, the traditional MVC H/Q strength ratio may not reflect the potential for dynamic knee joint stabilization during rapid match situations. Rather, the ability to rapidly activate the hamstrings relatively to the quadriceps is important. The ability to rapidly activate the muscle can be assessed in a standardized manner as rate of force development (RFD) during maximal voluntary static contraction (1).

In this study, we introduce an RFD H/Q strength ratio to reflect the ability to reach a given antagonist vs. agonist joint moment rapidly in explosive movements. Because it is not possible to investigate the RFD H/Q strength ratio during closed kinetic chain movements, such as sidecutting and landing, the isometric RFD H/Q strength ratio provides the

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Table 1. Demographics, RFD at 50 and 200 milliseconds from the onset of contraction, and MVC strength for Q and H of female and male players.*†

		Females		Males	
Age (y)		22.0 ± 2.4		24.5 ± 2.3	
Height (m)		1.68 ± 0.04		1.83 ± 0.07‡	
Weight (kg)		62.4 ± 6.0		78.4 ± 8.1‡	
BMI ($kg \cdot m^{-2}$)		22.2 ± 2.1		21.5 ± 2.3	
		H	Q	H	Q
RFD ($N \cdot m \cdot s^{-1} \cdot kg^{-1}$)	0–50 ms	10.2 ± 3.1	26.6 ± 6.9	13.0 ± 5.0§	35.0 ± 4.3‡
	0–200 ms	6.2 ± 1.2	14.5 ± 2.9	7.0 ± 1.0§	16.6 ± 1.7‖
MVC ($N \cdot m \cdot kg^{-1}$)		1.6 ± 0.3	3.7 ± 0.7	1.8 ± 0.2§	4.0 ± 0.5

*MVC = maximal voluntary contraction; RFD = rate of force development; BMI = body mass index; Q = quadriceps; H = hamstrings.
†Values are given as mean ± *SD*.
‡Men > women, $p < 0.01$.
§Tendency for men > women, $p = 0.07–0.12$
‖Men > women, $p < 0.05$.

best standardized estimate of the potential for dynamic knee joint stabilization. Furthermore, isometric RFD has been found to correlate with dynamic functional performance (15,27).

One of the most serious knee injuries in sports is rupture of the ACL. The relative incidence of noncontact ACL injury has been reported to be higher in female compared to in male athletes (6,22). Thus, potential gender differences in the ability for dynamic knee joint stabilization is an important aspect to considered in this study. In fact, female athletes have been found to rely more on their quadriceps muscles in situations of rapid anterior tibial translation, whereas male athletes rely more on their hamstring muscles for initial knee stabilization (14).

The aim of this study was to introduce and investigate the RFD H/Q strength ratio and to compare this with the traditional MVC H/Q strength ratio in male and female elite soccer players. We hypothesize that RFD H/Q strength ratio is lower in the very initial phase of muscle contraction compared to the MVC H/Q strength ratio. Further, we hypothesize that female compared to male elite soccer players have a lower RFD H/Q strength ratio in the very initial phase of contraction—predisposing for higher risk of ACL injury.

Methods

Experimental Approach to the Problem

This study evaluated the hypothesis that, when examining the very initial phase of muscle contraction, the ability to rapidly activate the hamstrings relatively to the quadriceps is reduced. Thus, the RFD H/Q strength ratio is hypothesized to be lower in the very initial phase of muscle contraction compared to the MVC H/Q strength ratio. Further, this study evaluated the hypothesis that female compared to male elite soccer players have a lower RFD H/Q strength ratio in the very initial phase of contraction. The ability to rapidly activate the hamstrings relatively to the quadriceps during explosive movements was assessed in a standardized manner as RFD during maximal voluntary static contraction. To obtain reproducibility of the introduced strength ratio, RFD H/Q strength ratio was measured in female athletes on 2 separate days. Further, The RFD H/Q strength ratio was obtained from female and male elite soccer players to determine whether a gender discrepancy existed in the very initial phase of contraction.

Subjects

A cross-sectional study was performed in Copenhagen, Denmark. Eleven female elite soccer players and 12 male elite soccer players, with no previous history of knee injury, were recruited from the top 2 divisions of Danish National Soccer

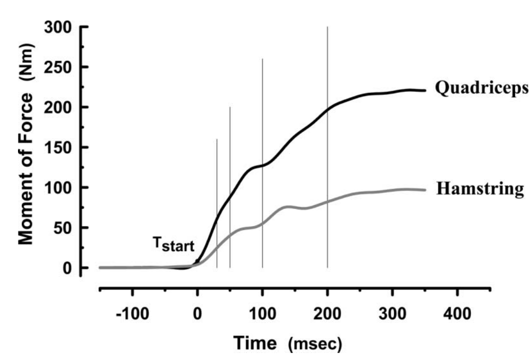

Figure 1. Moment–time curve for hamstring (gray line) and quadriceps (black line) muscles, respectively. Tstart represents onset of contraction. The vertical lines represent time point 30, 50, 100, and 200 milliseconds, respectively.

Table 2. Test–retest reliability for RFD H/Q strength ratio and MVC H/Q strength ratio. *

Test–retest reliability of H/Q strength ratio		
	ICC3.1	*p* Value
MVC H/Q ratio	0.751	0.043
RFD H/Q ratio	0.664–0.933	0.001–0.087

*MVC = maximal voluntary contraction; RFD = rate of force development; Q = quadriceps; H = hamstrings.

League. Training frequency was 5 ± 3 sessions per week. Subjects demographics are detailed in Table 1. The players were tested at end of match season for maximal (MVC) and explosive (RFD) quadriceps and hamstring muscle strength (Table 1). Test–retest reliability was performed separately in 8 comparable female players (mean age: 26 ± 2 years; mean height: 172 ± 6 cm; mean weight: 66.2 ± 3.5 kg) on 2 separate days (9 ± 4 days). The players were fully informed of all experimental procedures and risks before giving their written informed consent to participate. The study conforms to the code of ethics of the Declaration of Helsinki and was approved by the Ethics Committee of Copenhagen and Frederiksberg municipalities.

Procedures

Isometric MVC was performed for both hamstring and quadriceps, in the preferred kicking leg, using a KinCom dynamometer (KinCom, Kinetic Communicator, Chattecx Corp., Chattanooga, TN, USA) locked at a static anatomical knee joint angle of 70°. The reliability and validity of the dynamometer have previously been described in detail (11). The subjects were tested seated in a rigid chair with an 80° hip flexion and arms crossed. The hip and lower limb was fixated by straps attached to the chair and lever arm of the dynamometer, respectively. The rotational axis of the dynamometer was aligned to the lateral femoral epicondyl, and the lower leg was attached to the dynamometer lever arm above the medial malleolus. Measurements were preceded by a 15-minute warm-up on a Monark Cycle Ergometer (Monark Exercise AB, Sweden), followed by a number of submaximal contraction trials in the dynamometer. To secure an accurate assessment of maximal isometric strength, online visual feedback of the instantaneous dynamometer force was provided to the subjects on a computer screen (16). Instructions were given to the subjects to perform all contractions as fast and forceful as possible to obtain both maximal force and RFD (7). Trials with visible initial countermovement were excluded and an extra trial was performed. For each type of muscle contraction (i.e., isometric knee flexion and extension), 3 maximal trials were performed, separated by a rest period of 45 seconds. The dynamometer force signal was sampled at 1,000 Hz using an external A/D converter (dt2801-A, Data Translation, Marlboro, MA, USA) and stored on a stationary computer. During later off-line analysis the dynamometer signal was digitally lowpass filtered at 12 Hz, using a fourth-order zero phase lag Butterworth filter, converted to Newtons and multiplied by the individual lever arm length to calculate moment of force. All recorded moments were corrected for the effect of gravity on the lower limb (4,28). The MVC was defined as the highest peak torque value of 3 maximal attempts.

Contractile Rate of Force Development

From the maximal voluntary isometric contractions, the contractile RFD) was defined as the slope of the moment–time curve (i.e., ΔMoment/ΔTime, N·m·s^{-1}) in incrementing time periods of 0–10, 0–20, 0–30…0–250 milliseconds from the onset of contraction (5). Onset was defined as the time point where the moment exceeded baseline by 6.0 and 3.0 N·m for the quadriceps and hamstring, respectively (1,26) (Figure 1). Contractile RFD for any of the examined time period was determined as the highest obtained value in any of the 3 trials.

Definition of a Hamstring/Quadriceps Strength Ratio Based on Rate of Force Development and Maximal Voluntary Contraction

An H/Q strength ratio based on rapid isometric force development is introduced in this study. The RFD H/Q strength ratio was calculated by dividing hamstring RFD with quadriceps RFD in the examined time periods (i.e., 0–10, 0–20, 0–30…and 0–250 milliseconds, for example, hamstring RFD_{0-50} divided by quadriceps RFD_{0-50}, etc).

$$\text{RFD H/Q} = \frac{\text{Hamstring RFD}_i}{\text{Quadriceps RFD}_i},$$

where i denotes time from onset of contraction.

$$\text{MVC H/Q} = \frac{\text{Hamstring MVC}}{\text{Quadriceps MVC}}.$$

Statistical Analyses

Two-way analysis of variance was performed using the mixed procedure in SAS version 9.1 for Windows (SAS Institute, Cary, NC, USA). Factors included in the model for the RFD H/Q strength ratio were *gender* (male, female), *time* (0–10, 0–20,…,0–50 milliseconds from the onset), and *gender by time* interaction. When a significant main effect was found Bonferroni-corrected post hoc tests were performed to locate relevant differences. Test–retest reliability was assessed by intraclass correlation coefficients ($ICC_{3.1}$). An alpha level of 5% was chosen as statistically significant. All values are reported as mean ± *SD*.

Results

The RFD and MVC expressed per kilogram bodyweight were generally lower for women compared to for men (Table 1).

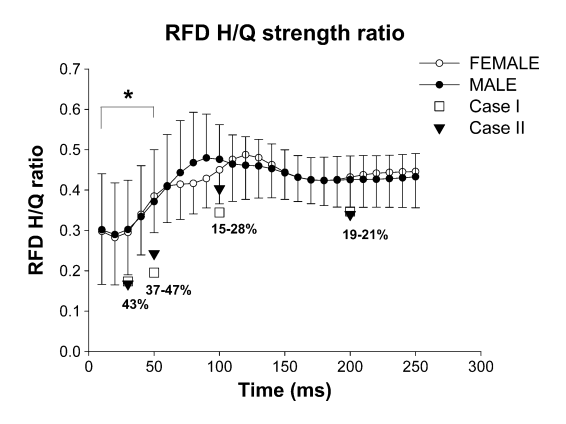

Figure 2. Rate of force development hamstring/quadriceps (RFD H/Q) strength ratio for female (○) and male (●), respectively. *Significantly different from latter time periods (≥60 milliseconds) and H/Q based on maximal voluntary contraction (MVC). Two female players (represented by a square and a triangle, respectively) were tested before having an anterior cruciate ligament (ACL) injury.

No difference existed in MVC H/Q strength ratio between male and female soccer players. The average MVC H/Q strength ratio for the collapsed group was 0.44 ± 0.07.

A high test–retest reliability (21) was found for the introduced RFD H/Q strength ratio in most of the examined time intervals (Table 2). Thus, the RFD H/Q strength ratio was generally as reliable as the traditional MVC H/Q strength ratio.

A priori hypotheses testing of main effects showed a significant *time* effect ($p < 0.0001$). Post hoc tests showed reduced RFD H/Q strength ratio during the very initial phase of contraction (up to 50 milliseconds from onset) compared to the MVC H/Q strength ratio ($p < 0.001$) (Figure 2). There were no significant *gender* ($p = 0.98$) or gender by time effect ($p = 0.93$), that is, there were no significant difference between male and female players.

Two female soccer players subsequently experienced ACL injury. Although these 2 players MVC H/Q ratio was not substantially different from the female group mean before injury (4% higher and 12% lower), they displayed a markedly lower RFD H/Q ratio in the very initial phase of contraction (37 and 48% lower at 50 milliseconds compared with the female group mean) (Figure2).

Discussion

This study shows reduced potential for knee joint stabilization during the very initial phase of voluntary muscle contraction in both genders. Thus, the RFD H/Q strength ratio compared to the MVC H/Q strength ratio was reduced in the initial 50 milliseconds in both female and male elite soccer players, suggesting impaired potential for agonistic hamstring protection of the ACL. This could have important clinical relevance because the estimated time of ACL injury, based on video analysis of 39 cases, ranges from 17 to 50 milliseconds after initial ground contact (18). In contrast, ACL injuries rarely ever occur during heavy controlled lifts, for example, during power lifting or weight lifting, where the time period is sufficient for the hamstrings to develop appropriate levels of force to protect the ACL. Furthermore, studies have shown that during a sidecutting maneuver where most noncontact ACL injuries are observed (23), neuromuscular activation of the hamstrings are only 30–50% of maximum activation (at MVC) (25,30), supporting that contractile hamstring RFD rather than maximal strength is an important parameter for the potential of muscle stabilization at the knee joint.

It has been proposed that evaluation of knee joint function by the use of isokinetic dynamometry should comprise data on dynamic or static MVC H/Q strength ratios combined with data on absolute muscle strength (2–4). Neither of these takes into account the rapid strength capacity in the very initial phase of contraction. The high test–retest reliability of the introduced RFD H/Q strength ratio indicates that changes over time, for example, during specific preventative hamstring training, can be tracked with a high level of accuracy. This could potentially make the method a relevant tool in standardized clinical evaluation of the knee joint agonist–antagonist relationship. It is important to emphasize that even though the presence of low MVC H/Q strength ratios has been suggested to be associated with an increased risk of knee injuries (17,29) and hamstring muscle injury (9), such a relationship has not yet actually been empirically established.

In contrast to our hypothesis, no gender differences in RFD H/Q or MVC H/Q strength ratios were observed in the present soccer players. At the time of testing, all players were uninjured and with no previous history of knee injuries. In this study, 2 female soccer players experienced an ACL injury the following year after testing. Although the MVC H/Q strength ratio of these 2 players was not substantially different from that of the female group mean before injury, they displayed a markedly low RFD H/Q strength ratio in the very initial phase of contraction at the time of testing. Obviously, no conclusions can be made based on only 2 cases. However, it could be speculated that the initial phase (i.e., < 50 milliseconds) of the RFD H/Q time curve may be used to identify players with a potentially higher risk for knee injury. Further, the 2 ACL cases of this study suggest that the RFD H/Q strength ratio more so than the MVC H/Q strength ratio is important for dynamic knee joint stabilization during rapid soccer movements. However, large prospective studies should be performed to confirm the present findings before recommendations of the present method as a clinical screening tool can be made.

Practical Applications

This study shows reduced potential for knee joint stabilization during the very initial phase of voluntary muscle contraction, suggesting impaired potential for agonistic hamstring protection of the ACL. The high reliability of the introduced RFD H/Q strength ratio potentially makes the method a relevant tool in standardized clinical evaluation of the knee joint agonist–antagonist relationship. Finally, high-intensity resistance training – aiming at increasing hamstring RFD in the early phase of contraction – may be essential in the prevention of ACL injury.

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